

Programming Using C++ Level 1

Output

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Outout



Definition

◆ Output means displaying information from a C++ program on the console screen.

Explanation

-  ``cout`` is used to print text on the screen.
-  The ``<<`` operator sends the text to ``cout``.
-  Text inside ``" "`` is displayed exactly as written.
-  ``endl`` moves the cursor to the ****next line**** after printing.

Steps in main()

1. ``cout`` → Prints the required information on the console.
2. ``<<`` → Sends the text to ``cout``.
3. ``endl`` → Moves the cursor to the next line.
4. The next ``cout`` → Prints the next piece of information.

 Result:

The information appears on the screen line by line

💡 Example

```
cout << "*****" << endl;  
cout << "Name: Sara Alterkawi." << endl;  
cout << "Age: 43 Years." << endl;  
cout << "City: Perstorp." << endl;  
cout << "Country: Sweden." << endl;  
cout << "*****" << endl;
```

Expected Output

Name: Sara Alterkawi.

Age: 43 Years.

City: Perstorp.

Country: Sweden.

Object 4 $\rightarrow$ x = 4

Note

- ◆ A default constructor is required when allocating arrays of objects using `new[]`
- ◆ This is why a dummy constructor is defined in the class

⊘ Warning

- ◆ Do not forget endl if you want each output to appear on a separate line.
- ◆ Without endl, multiple outputs may appear on the ****same line****.

```
cout << "Sara";  
cout << "Alterkawi";
```

➡ Result:

SaraAlterkawi

Comparison

With endl

- ✓ Each message appears on a new line.
- ✓ Makes the output easier to read.

```
cout << "Sara" << endl;  
cout << "Alterkawi";
```

➡ Result:

Sara
Alterkawi

Without endl

- ✓ Messages appear continuously on the same line.
- ✓ No automatic new line is created.

```
cout << "Sara";  
cout << "Alterkawi";
```

➡ Result:

SaraAlterkawi

✦✦ Summary

- ✓ `cout` is the main tool we use to display **output** in C++.
- ✓ `endl` is used when we want to move to a **new line** after printing.
- ✓ The output is displayed **exactly as written inside the quotation marks**.

Thank you 😊